Chemical Research Note: Aspirin Hydrolysis Study

This document describes a small laboratory investigation of acetylsalicylic acid, commonly known as aspirin. The research objective was to observe hydrolysis under mild alkaline conditions and identify the main products formed during the reaction.

Aspirin has the molecular formula C9H8O4 and a molar mass of 180.16 g/mol. In the presence of sodium hydroxide and water, aspirin can hydrolyze to form salicylic acid and acetate. The reaction was monitored at room temperature using pH measurements and thin-layer chromatography.

Experimental setup: 0.50 g of aspirin was dissolved in 25 mL of ethanol-water solution. A dilute NaOH solution was added dropwise while stirring. Samples were collected after 5, 15, 30, and 60 minutes. The expected compounds were aspirin, salicylic acid, sodium acetate, ethanol, and water.

Observed result: the aspirin spot decreased over time, while a new spot consistent with salicylic acid appeared after 15 minutes. The solution became slightly basic after addition of NaOH. These observations support the interpretation that aspirin underwent base-assisted hydrolysis.

Example facts for extraction:
- Aspirin has molecular formula C9H8O4.
- Aspirin hydrolyzes to salicylic acid and acetate.
- Sodium hydroxide acts as a base in the reaction.
- Thin-layer chromatography was used to monitor the reaction.
- Salicylic acid was detected after 15 minutes.